

The Communications Revolution, 1850–1940

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From his tri-fold exhibit “The Communications Revolution” · Part 2 of 3

1850 -1870

The Telegraph Age

- **Expansion:** Telegraph lines became commonplace, connecting cities and supporting business with speed.
- **Civil War Use:** Expansion of the telegraph network makes President Lincoln to implement and utilize to gain situational awareness of the battlefield and issue orders during the Civil War. The telegraph played a major role in the success of the Union and culminating in the end to the Civil War.
- **Semaphore Flag Signaling**: Developed and implemented by BG Myers during the Civil War and complemented Telegraph Use where telegraph lines were unavailable (1860s)
- **Transatlantic Cables (1858, 1866):** The first successful, lasting transatlantic cable was completed, connecting Europe and America, making communication nearly instant.
- **Commercial Growth:** Western Union was established and monopolized the US industry, overseeing a vast network of hundreds of thousands of miles of telegraph lines.
- **Professional Use:** The US Department of State adopted the telegraph for diplomacy (1866).

Key Themes in 1870s Telephony Research

- **The Harmonic Telegraph Quest (Early 1870s):** Inventors, including Alexander Graham Bell and Elisha Gray, initially aimed to create a "multiple telegraph" to send several messages simultaneously over one wire.
- **Shift to Voice Transmission (1874-1875):** Bell shifted focus from multiple notes to sending vocal tones via variable resistance, realizing electric current could reproduce voice.
- **The 1876 Breakthrough:** On March 7, 1876, Bell was awarded the "master patent" for the electric telephone, which used a diaphragm and electromagnetic signaling to transmit voice.
- **The Patent Race:** Both Bell and Elisha Gray independently developed prototypes and filed documents within hours of each other in February/March 1876, sparking legal battles that Bell eventually won.
- **Telephone Improvements**: In 1878 Thomas Edison patented the carbon transmitter (microphone) that enhanced sound quality for Bell's Telephone

Audio Recording (Phonography)

- **1877**: Invention of the Phonograph
  - Thomas Edison invented the phonograph, the first machine capable of recording and reproducing sound (initially on tinfoil)
- **1878**: Demonstration ; The device was perfected at his Menlo Park, NJ laboratory introducing the ability to store audio information

Improvements in Information Technology

- **1870's**: Printing Press Improvements: Technical advancements allowed for faster printing, supporting the rapid spread of news through mass circulation newspapers
- **1880: Photophone Experimentation**: Alexander Graham Bell and Charles Tainer invented the photophone in 1880 which transmitted voice via light beams = a precursor to fiber optics

1880-1910

Wireless Telegraphy/Radio (1890s–1900s)

- Guglielmo Marconi began experimenting with radio in 1894, transmitting signals over longer distances.
- Nikola Tesla develops patent for Tuned Circuit and Antenna forming the basis for wireless receive and transmit capabilities
- Nikola Tesla demonstrates the first use of remote control by inventing and demonstrating a remote control boat at Madison Square Garden, NYC
- In 1901, Marconi transmitted the first radio signal (the letter 'S') across the Atlantic Ocean.
- In 1908, Lee de Forest broadcasted music from the Eiffel Tower, demonstrating the potential for audio transmission.

Telephony Expansion (1880s–1910)

- Following the 1876 patent, telephone exchanges began opening, with rapid growth in the 1880s.
- By 1900, there were roughly 600,000 phones in the Bell system, rising to 5.8 million by 1910.
- The introduction of the multiplex switchboard in 1887 allowed for faster, more efficient call routing.

Mass Communication and Office Technology

- The widespread adoption of typewriters revolutionized business correspondence.
- Advancements in printing technology made mass distribution of newspapers and information faster.

Early Recordings:

- Thomas Edison's phonograph, patented late in the 1870s, was refined in the 1880s/90s to become a popular entertainment medium.

Note: These innovations allowed for the rapid growth of large-scale, efficient, and direct communication networks that formed the basis for modern, interconnected global systems.

1910-1920

Wireless Telegraphy and Radio Advancement

- **Wartime Radio Expansion:** World War I forced rapid advancements in radio technology, moving it beyond specialized nautical use to critical military application
- **The “Regenerative” Circuit (1913)** is invented and tested by Edwin Armstrong at Marconi Wireless Station in Belmar NJ. The circuit significantly increased the amplification of radio signals by 5000 times using feedback
- **The "Superheterodyne" Circuit (1918):** Inventor Edwin Armstrong developed the superheterodyne receiver, which drastically improved the sensitivity and selectivity of radio receivers, setting the standard for future radio technology.
- **Voice Transmission:** Following the first long-range voice transmission in 1906, voice broadcasting continued to mature throughout the 1910s, laying the foundation for commercial radio.

Telephone Network Expansion

- **Transcontinental Line (1915):** The first official transcontinental telephone line was opened, allowing direct voice communication across the United States.
- **Rapid Adoption:** Telephone usage grew rapidly, with the number of phones in the Bell system reaching millions, becoming an essential tool for business and personal communication.

Mass Media and Information

- **Propaganda & Print:** Newspapers and early film industries were utilized to spread government propaganda during WWI, establishing new models for mass communication.
- **Technological Roles:** The era created new professional positions, including thousands of new typists and telegraphers, supporting the industrialization of information.

Early Experimental Broadcasting

- **Amateur to Commercial:** While 1920 is considered the official start of commercial radio with KDKA, 1916 marked a critical shift when radio enthusiasts (like Frank Conrad) began transmitting regular, scheduled music programs.

Note: This decade transformed the “job market” with the rise of new positions trained typists and telegraphers to handle the accelerating flow of information . Additionally, this era laid the foundation for the “Golden Age of Radio” that began shortly after the 1920s

1920-1940

Wartime/Radar and Radio Technology

Frequency Modulation (FM)

- Major Edwin Armstrong invents Frequency Modulation (FM) circuit and patents in 1933. FM utilizes a limiter and discriminator that eliminates static noise making communications clear a crisp. Armstrong gave US Government patent rights during WWII resulting fielding of reliable clear communications equipment

RADAR (Radio and Direction Finding)

- **1935/1939:** Sir Robert Watson-Watt developed the first practical radar system (1935), leading to the British "Chain Home" radar network to detect aircraft/sea vessels by 1939.
- **1940:** The resonant-cavity magnetron was perfected, significantly increasing radar power and allowing for smaller, microwave radar systems

NOTE: 1941: The SCR 271 Radar was fielded in Pearl Harbor by the US Army. It detected aircraft but was not identified as enemy aircraft

Television Development (1920s-1930s)

- **1925/1926:** John Logie Baird transmitted the first recognized television signal (1925) and publicly demonstrated the device (1926).
- **1930s:** Experimental television broadcasts began, with regular, limited broadcasting schedules starting in some cities towards the end of the 1930s.

Advancements in Telephony (1920s-1930s)

- **1924:** Bell Laboratories developed the first two-way, voice-based radio telephone, enabling crucial long-distance maritime and aviation communication.
- **1930s:** The telephone shifted from a luxury to a critical business and personal tool, improving communication between companies and customers.